

FlatCFM: Conditional Flow Matching in Flat Latent Space for Single Cell Perturbation Modeling

Project Proposal

David Crair
david.crair@yale.edu

Advisor: Smita Krishnaswamy
smita.krishnaswamy@yale.edu

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1 Introduction

Brief overview of your project and its purpose.

Transcriptomic data is valuable to both biology and drug discovery to see heterogeneous effects of drug and gene perturbations, but current models often fail to generalize known perturbation effects to unseen biological contexts (e.g. cell lines or patients). The ability to generalize to underrepresented contexts reduces wet lab burden and allows for transfer of knowledge from contexts that are less challenging to produce data for to contexts that are infeasible or otherwise prohibitively expensive to generate data in.

This project proposes a novel combination of existing approaches to modeling single-cell transcriptomic data for perturbation prediction. By flattening the latent space of an autoencoder and performing flow matching in this latent space, the straight-line flows that flow matching trains on will correspond to geodesics on the gene expression manifold. This will result in better model performance by ensuring that interpolated cellular trajectories respect the underlying biological manifold structure. Even though the task is not trajectory inference as in Sun et al. or Palma et al., I theorize that this flattening will still be beneficial for the perturbation prediction task as sampled linear interpolations between source and target distributions as done in flow matching will correspond to logical biological states on the manifold.

2 Background/Related Works

Summary of existing research or technologies related to your project.

Over the past 5 years, many different methods have appeared for the task of perturbation prediction on single cell data. Initial methods involved GNNs and compositions of embeddings in the latent space of autoencoders, whereas more recent methods have increased in complexity, including flow matching, transformers, and diffusion. Unlike language modeling, this subfield has yet to mature or standardize on a common framework for either model architecture or evaluation metrics and tasks. Below are some of the major paper and related methods related to this project.

CPA [Lotfollahi et al.] is the first major paper to tackle the problem of predicting perturbation effects in single cell data for a composition of perturbations and completely unseen perturbations. CPA learns independent embeddings for the cell type, perturbation, and other covariates so that they can be combined to predict the effect of new combinations of perturbations and conditions.

GEARS [Roohani et al.] is a GNN-based approach to predict perturbation effects which combines GNN-generated embeddings of perturbations and genes and passes their composition through different MLP layers.

CellFlow [Klein et al.] performs optimal transport conditional flow matching (OT-CFM) directly on the gene expression space, using a condition embedding formed by aggregating covariate encodings to predict the effects of gene knockouts, drug treatments, and development, on a wide range of datasets. The network outputs velocity vectors at any point in the gene expression space based on the condition embedding and a numerical ODE solver integrates the vector field from the starting cell ($t = 0$) to the ending cell ($t = 1$) to make a prediction.

GAGA [Sun et al.] regularizes an autoencoder to match geodesic distances in the ambient space with euclidean distances in the latent space and additionally computes a warped pullback metric to define a Riemannian metric off the manifold. This can be used for a variety of tasks, including uniform data generation, transport along geodesics, and interpolation of cellular trajectories, but the methods from GAGA have not been used to do perturbation prediction for single cell data.

FlatVI [Palma et al.] is similar to GAGA in that it also regularizes an autoencoder such that straight lines in the latent space approximate geodesics in the gene expression space. FlatVI flattens the latent space of the autoencoder by regularizing the pullback metric of the decoder to be a scaled version of the identity matrix. FlatVI is more specific to single cell data, as the decoder defines parameters for a Negative Binomial distribution, which is specifically chosen for

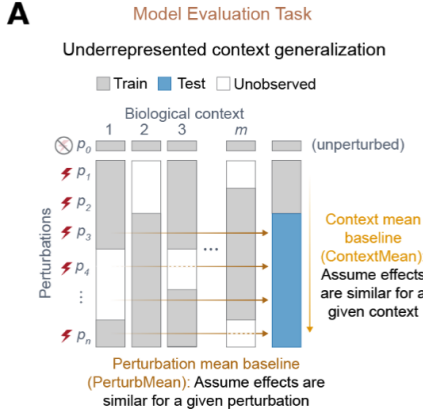


Figure 1: Underrepresented context generalization task, from Adduri et al.

single cell data. Also like GAGA, FlatVI has not been used for perturbation prediction.

scGPT and scPPDM Most recently, diffusion models and transformer-based models have started to be used in this space. scGPT [Cui et al.] is a transformer-based foundational model which generates embeddings of cells from scRNA-seq data and can be used to predict perturbation responses, among other downstream tasks. scPPDM [Liang et al.] performs perturbation response prediction

3 Problem Description

Clearly state the problem you aim to solve.

The main problem I aim to solve in this project is predicting the heterogeneous single-cell transcriptional response in underrepresented contexts. In this task, there are a set of single drug, gene, or cytokine perturbations across different contexts, such as cell line or patient. In the training data, we have all available post-perturbation gene expression data across a set of training context as well as the control (non-perturbed) gene expression and a subset of perturbations in the test contexts. The goal is to generalize to predict the post-perturbation gene expression in the held-out perturbations of the test (underrepresented) contexts. There is very high overlap of perturbations across contexts, with each perturbation appearing in over 90% of training contexts consistently across datasets. Table 1, taken from Adduri et al. shows a visual representation of this task, which is one of the most common perturbation prediction tasks.

The metrics that will be used to evaluate the performance of the model relative to the baseline and other models are:

- Squared Wasserstein Distance (2-WD) [Klein et al.]
- Mean Maximum Discrepancy (MMD) [Klein et al.]
- Rank [Wu et al.]
- Transposed Rank [Wu et al.]
- Cosine logFC [Wu et al.]
- E-distance

The main datasets I will use for this project are the OP3 perturbation benchmark from the 2023 NeurIPS competition [Szałata et al.], a PBMC dataset with drug perturbations, Parse-PBMC [<https://www.parsebiosciences.com/datasets/10-million-human-pbmcs-in-a-single-experiment/>], a PBMC dataset with cytokine perturbations, and Replogle-Nadig, a genetic perturbation dataset with 4 cell lines [Nadig et al.].

4 Planned Approaches

Describe your proposed methods or solutions.

To flatten the latent space of the autoencoder, I plan to compare using a method from GAGA [Sun et al.] and a method from FlatVI [Palma et al.]. Sun et al. matches geodesic distances in the gene expression space with euclidean distances in the latent space of an autoencoder, while Palma et al. regularize the pullback metric of the decoder of the autoencoder to match a scaled Identity matrix and additionally decode from a statistical manifold using a negative binomial distribution.

In the latent space of the autoencoder, I plan to use the same flow matching approach as in CellFlow [Klein et al.], which uses the OT-CFM framework and injects aggregated covariate encodings as a condition embedding to direct the flow. For the underrepresented context generalization task, we need condition embeddings for the perturbations, the perturbation covariates (dose, timing) and the sample covariates (cell line, patient, batch) as in Klein et al.. I plan to use Deep Sets-style aggregation of the encodings (as in Klein et al.), which transforms each with an MLP, sums, then transforms the sum with another MLP [Buterez et al.].

To encode drugs and genes, I plan to use Morgan fingerprints and ESM2 embeddings respectively. This will allow for the model to do single-gene and multi-gene completely unseen drug and gene perturbation prediction. I may also try using existing alternative molecular encodings, such as MolFormer [Ross et al.], ChemBerta [Singh et al.], or CDDD [Winter et al.]. I plan to use learned embeddings for the perturbation covariates and sample covariates and may try learned embeddings for the perturbations as well. When applicable, I experiment with using existing cell line encodings, such as DREEP [Pellecchia et al.].

5 Schedules/Timelines

Include dates or weekly milestones to track your progress and stay on schedule.

Weeks 1-3 (1/12-2/1)

- Literature review, write project proposal, plan project, brainstorm ideas.
- Submit project proposal (due 1/29 at Noon)

Weeks 4-6 (2/2-2/22)

- Give project proposal presentation (2/13)
- Implement covariate encodings and aggregation methods
- Implement OT-CFM with condition embedding in PyTorch
- Implement Geodesic regularization and decoder pullback regularization in PyTorch and test on both toy datasets and scRNA-seq data

Weeks 7-10 (2/23-3/22)

- Implement all baselines
- Implement all metrics
- Write midterm progress report
- Submit midterm progress report (due 3/6 at Noon)
- Benchmark baseline methods on core datasets

Weeks 11-14 (3/23-4/19)

- Give midterm progress report presentation (4/10)
- Run hyperparameter search
- Create figures for final report
- Write final report
- Make final poster

Weeks 15-16 (4/20-5/30)

- Submit final report to advisor (due 4/20 at Noon)
- Finish final poster
- Submit final poster (due 4/22)
- Revise final report based on feedback
- Submit final report (due 4/30 at Noon)

6 List of Deliverables

Identify specific outcomes (code, prototypes, poster, reports, etc.).

Code

- OT-CFM implementation
- Geodesic regularization loss implementation
- Decoder pullback regularization implementation

- Baseline methods: linear additive, latent additive, no effect, perturb mean, context mean, decoder only [Wu et al.] [Adduri et al.]

Report

- **Figure:** Comparison of two autoencoder regularization methods on toy dataset(s) (e.g. swiss roll, s-curve)
- **Table:** Comparison of performance on small dataset at different autoencoder regularization strengths across both autoencoder regularizations.
- **Figure or Table:** Visual or numeric component showing that velocity vectors in the latent space with regularization are straighter than without regularization (e.g. table showing path length in ambient space for example paths)
- **Figure:** Plot showing correlation between euclidean distances in the latent space and geodesic distances in the gene space
- **Figure:** Plot comparing distribution in PCA space of control vs predicted vs ground truth for perturbation(s)

Misc

- project proposal (this document)
- midterm progress report
- final report
- final poster
- GitHub repo

7 References

List all sources cited in your proposal.

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